

Chapter 1

Scalar Linear Differential Equations

$$(1) \quad x^{(n)} + a_1(t) x^{(n-1)} + a_2(t) x^{(n-2)} + \dots + a_{n-1}(t) x' + a_n(t) x = f(t)$$

$a_1, \dots, a_n, f \in C(J)$, $J \subset \mathbb{R}$
continuous functions on J

A solution of (1) is a function $\varphi \in C^n(J)$ that satisfies (1),
i.e. $\varphi^{(n)}(t) + a_1(t) \varphi^{(n-1)}(t) + \dots + a_n(t) \varphi(t) = f(t)$, for all $t \in J$.

$a_1, \dots, a_n \Rightarrow$ coefficients

$f \Rightarrow$ non-homogeneous part of (1)

If $a_1, \dots, a_n$ are constant $\Rightarrow$ L.D.E. with CC

If $f \equiv 0 \Rightarrow$ linear homogeneous. Otherwise, it's ^{linear} nonhomogeneous.

Let $t_0 \in J$ and $\eta_1, \dots, \eta_n \in \mathbb{R}$ be given. We have that the IVP

$$(2) \quad x^{(n)} + a_1(t) x^{(n-1)} + \dots + a_n(t) x = f(t), \quad x(t_0) = \eta_1, \dots, x^{(n-1)}(t_0) = \eta_n.$$

has a unique solution which is defined on the whole J .

Take (3): $x^{(n)} + a_1(t) x^{(n-1)} + \dots + a_{n-1}(t) x' + a_n(t) x = 0$

When eq. (1) is lin. homom., we say (3) is the
LHDE associated to it.

§ The Linearity Principle: If $x_1, x_2, \dots, x_m$ are solutions of (3),
then for $\forall \alpha_1, \dots, \alpha_m \in \mathbb{R}$, we have that $(\alpha_1 x_1 + \dots + \alpha_m x_m)$
is also a solution of (3).

!! The Fundamental Theorem for linear homogeneous $\Delta \in \Delta$.

~ The set of solutions of the linear hom. m -th order eq. (3) is a linear space of dimension m . Thus, $\exists x_1, \dots, x_m$ lin. ind. solutions of (3) and its gen. sol. can be described as

$$x = c_1 x_1 + \dots + c_m x_m; \quad c_1, \dots, c_m \in \mathbb{R}$$

!! The Fundamental Theorem for linear nonhomogeneous DEs.

~ Let x_h be the general solution of the linear hom. DE associated to (1) and let x_p be a particular solution of (1). Then the general solution of (1) is

$$x = x_h + x_p$$

! The general solution of a first order L.D.E. !

$$\text{Take } x' + a(t)x = f(t) \quad (4),$$

where $a, f \in C(J)$ and write also the linear hom. eq. associated

$$x' + a(t)x = 0 \quad (5).$$

Let $t_0 \in J$ be fixed and denote by A a primitive of a , i.e.

$$A(t) = \int_{t_0}^t a(s) ds.$$

$\Rightarrow$ We have that $x_1 = e^{-A(t)}$ is a solution of (5).

Hence the gen. sol. of this DE is $x = c \cdot e^{-A(t)}$, $c \in \mathbb{R}$.

By the Lagrange Method, we can find a particular solution for (4).

$$x_p = \int_{t_0}^t e^{-A(t)+A(s)} \cdot f(s) ds.$$

2 Methods to find the solutions of a given equation 2

- The algorithm for determining the general solution for the m^{th} order LHS with cc.

Step 1: We attach to the LHS

$$x^{(m)} + a_1 x^{(m-1)} + \dots + a_{m-1} x' + a_m x = 0$$

the characteristic equation: $\lambda^m + a_1 \lambda^{m-1} + \dots + a_{m-1} \lambda + a_m = 0$ (*)

Step 2: Determine the roots of the char. eq. (*)

Step 3: The functions $x_1(t), \dots, x_m(t)$ are attached to each root $\lambda_1, \dots, \lambda_m$ in the following way:

a) If $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ s.t. $\lambda_1 \neq \lambda_2 \neq \dots \neq \lambda_m$, then
 $x_1(t) = e^{\lambda_1 t}, x_2(t) = e^{\lambda_2 t}, \dots, x_m(t) = e^{\lambda_m t}$

b) If $\lambda_1, \dots, \lambda_m \in \mathbb{R}$ s.t. $\lambda_1 = \lambda_2 = \dots = \lambda_m$, then
 $x_1(t) = e^{\lambda_1 t}, x_2(t) = t e^{\lambda_1 t}, \dots, x_m(t) = t^{m-1} e^{\lambda_1 t}$

c) If $\lambda_1, \dots, \lambda_m \in \mathbb{C}$, $\lambda_1, \dots, \lambda_m = \alpha \pm i\beta$, then
 $x_1(t) = e^{\alpha t} \cos(\beta t), x_2(t) = e^{\alpha t} \sin(\beta t),$
 $x_3(t) = t e^{\alpha t} \cos(\beta t), x_4(t) = t e^{\alpha t} \sin(\beta t) \dots$
 $\dots x_{2m-1}(t) = t^{m-1} e^{\alpha t} \cos(\beta t), x_{2m}(t) = t^{m-1} e^{\alpha t} \sin(\beta t)$

- Separation of Variables •

Example: $x' - 3tx = 0$

$$\frac{dx}{dt} = 3tx$$

$$\frac{dx}{x} = 3t dt$$

$$\int \frac{1}{x} dx = \int 3t dt$$

$$\ln x = 3 \frac{t^2}{2} + C \quad \uparrow e$$

$$x = e^{\frac{3}{2}t^2} \cdot e^C$$

• Integrating factor •

$$x'(t) + a(t) \cdot x(t) = f(t) \quad | \cdot e^{-A(t)}$$

We choose $\mu(t) = e^{-A(t)} = e^{-\int_{t_0}^t a(s) ds}$ as the integrating factor

$$x' \cdot e^{-A(t)} + a(t) \cdot x \cdot e^{-A(t)} = f(t) \cdot e^{-A(t)}$$

$$\left[x \cdot e^{-A(t)} \right]' = f(t) \cdot e^{-A(t)} \quad | \int_{t_0}^t ds$$

$$x \cdot e^{-A(t)} = \int_{t_0}^t f(s) \cdot e^{-A(s)} ds + c \quad | \cdot e^{A(t)}$$

$$x(t) = e^{A(t)} + e^{A(t)} \cdot \int_{t_0}^t f(s) \cdot e^{-A(s)} ds, \quad \forall c \in \mathbb{R}$$

Example: $x' - 2x = 4 - t$ ↖ ~~2x~~ a

$$\mu(t) = e^{\int -2 dt} = e^{-\int 2 dt} = e^{-2t}$$

$$x' - 2x = 4 - t \quad | \cdot e^{-2t}$$

$$x' \cdot e^{-2t} - 2x \cdot e^{-2t} = (4 - t) e^{-2t}$$

$$(x \cdot e^{-2t})' = (4 - t) e^{-2t} \quad | \int dt$$

$$x \cdot e^{-2t} = \int (4 - t) e^{-2t} dt$$

$$\underbrace{\int 4 e^{-2t} dt}_{(-2 \cdot e^{-2t})'} - \underbrace{\int t \cdot e^{-2t} dt}$$

$$= -2 \int t \cdot \left(-\frac{1}{2} e^{-2t} \right)' dt =$$

$$= 2 \left[-\frac{1}{2} \cdot t \cdot e^{-2t} - \int -\frac{1}{2} e^{-2t} dt \right] =$$

$$= t \cdot e^{-2t} - \int e^{-2t} dt =$$

$$= \dots = t \cdot e^{-2t} - \frac{1}{2} e^{-2t} + \frac{1}{4} e^{-2t} + c \quad | \cdot e^{2t}$$

$$\Rightarrow x(t) = -2 + \frac{t}{2} + \frac{1}{4} + c \cdot e^{2t}$$

Chapter 4

The Dynamical System generated by a Differential Equation

We consider differential equations of the form

$$\dot{x} = f(x)$$

Theorem

Let $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ and $\eta \in \mathbb{R}^n$. Then the IVP

$$(2) \quad \begin{aligned} \dot{x} &= f(x) \\ x(0) &= \eta \end{aligned}$$

has a unique solution defined on an open (maximal) interval $J_\eta = (\alpha_\eta, \omega_\eta) \subset \mathbb{R}$. Denote this solution by $\phi(\cdot, \eta)$.

If $\phi(\cdot, \eta)$ is $\begin{cases} \text{bounded, then } J_\eta = \mathbb{R} \\ \text{bounded to the right, then } \omega_\eta = \infty \\ \text{bounded to the left, then } \alpha_\eta = -\infty \end{cases}$

The map ϕ of 2 variables t and η defined in the previous theorem is called the flow of the dyn. sys. gen. by (1). Some important properties of this map are

- $\rightarrow \phi(0, \eta) = \eta$
- $\rightarrow \phi(t+s, \eta) = \phi(t, \phi(s, \eta))$
- $\rightarrow \phi$ is continuous with respect to η .

When working with the flow, η = the initial state, where $\phi(t, \eta)$ = the state at time t .

We say that $\eta^* \in \mathbb{R}^m$ is an equilibrium point of the dym. sys. gen. by (1) when $f(t, \eta^*) = \eta^*$, $\forall t \in \mathbb{R}$.

→ The eq. points can be found by solving the equation $f(x) = 0$.

The orbit of the initial state η is:

$$\mathcal{J}(\eta) = \{ \varphi(t, \eta) : t \in \mathcal{I}_\eta \}$$

The positive orbit of the initial state η is:

$$\mathcal{J}^+(\eta) = \{ \varphi(t, \eta) : t \in \mathcal{I}_\eta, t > 0 \}$$

The negative orbit of the initial state η is:

$$\mathcal{J}^-(\eta) = \{ \varphi(t, \eta) : t \in \mathcal{I}_\eta, t < 0 \}$$

! The Linearization Method !

Let η^* be an eq. solution of $\dot{x} = f(x)$.

< If $f'(\eta^*) < 0$, then η^* is an attractor,
If $f'(\eta^*) > 0$, then η^* is a repeller.

If η^* is an attractor, its basin of attraction is

$$A_{\eta^*} = \{ \eta \in \mathbb{R}^m : \lim_{t \rightarrow \infty} \varphi(t, \eta) = \eta^* \}$$

When $A_{\eta^*} = \mathbb{R}^m$, we say that η^* is a global attractor.

Steps for representing the phase portrait of any scalar dynamical system $\dot{x} = f(x)$.

Step 1 Find all the equilibria. That is, solve $f(x) = 0$.

Step 2 Represent the equilibria on the state space, $\mathbb{R}$.

Step 3 Determine the sign of f on each orbit. If the sign is $+$, the arrow must indicate to the right, while if the sign is $-$, the arrow must indicate to the left.

~ Stability of the equilibria of dynamical systems ~

Roughly speaking, an eq. point η^* is stable if all solutions starting near η^* stay nearby.

In addition, if nearby solutions tend to η^* as $t \rightarrow \infty$, then η^* is asymptotically stable.

$$\boxed{\dot{x} = f(x) \quad (1)}$$

! An eq. point η^* is said to be asymptotically stable if it's both stable and attractor. !

We consider $\dot{x} = Ax \quad (6)$, where the matrix $A \in \mathcal{M}_n(\mathbb{R})$ is called the matrix of the coefficients of the linear system (6). We assume that $\det A \neq 0$, s.t. the only eq. point of (6) is $\eta^* = 0$ (here 0 denotes the null vector from $\mathbb{R}^n$).

→ We say that the lin. sys. (6) is stable / unstable / asymptotically stable when its eq. point at the origin has this quality.



Denote by $\sigma(A)$ the set of all eigenvalues $\lambda \in \mathbb{C}$ of matrix A .

!! remember !!

↳ eigenvalues are computed by: $\det(A - \lambda I_m) = 0$

and eigenvectors: $(A - \lambda I_m)v = 0$

If $\operatorname{Re}(\lambda) < 0$ for $\forall \lambda \in \sigma(A) \Rightarrow \dot{x} = Ax$ is ^{an} asympt. stable system.

If $\exists \lambda \in \sigma(A)$ s.t. $\operatorname{Re}(\lambda) > 0$, the linear system $\dot{x} = Ax$ is unstable.

An eq. point η^* of (1) is said to be hyperbolic when $\operatorname{Re}(\lambda) \neq 0$ for any eigenvalue λ of the Jacobian matrix $J f(\eta^*)$.

!! Theorem !!

Let η^* be a hyperbolic eq. point η^* of (1). We have that η^* is asymptotically stable/unstable iff the linear system $\dot{x} = J f(\eta^*)x$ has the same quality.

Corollary 1: Let $m=1$ and η^* be an eq. point of $\dot{x} = f(x)$.

If $f'(\eta^*) < 0 \Rightarrow \eta^*$ is asymptotically stable.

If $f'(\eta^*) > 0 \Rightarrow \eta^*$ is unstable.

2 Phase portraits of planar systems 2

We consider $\dot{x} = Ax$, where $A \in M_2(\mathbb{R})$, with $\det A \neq 0$.

In this case, the state space is $\mathbb{R}^2$ and the orbits are curves.

Denote by $\lambda_1, \lambda_2 \in \mathbb{C}$ the 2 eigenvalues of A . In the next definition, the eq. point at the origin is classified as node / focus / center / saddle, depending on the eigenvalues of A .

The eq. point $\eta^* = 0$ of the linear planar system $\dot{x} = Ax$ is a:

NODE if $\lambda_1 \leq \lambda_2 < 0$ or $0 < \lambda_1 \leq \lambda_2$

SADDLE if $\lambda_1 < 0 < \lambda_2$

FOCUS if $\lambda_{1,2} = \alpha \pm i\beta$, with $\beta \neq 0$ and $\alpha \neq 0$

CENTER if $\lambda_{1,2} = \pm i\beta$, with $\beta \neq 0$.

! Remark !

→ A node can be $\left\{ \begin{array}{l} \text{asymptotically stable: when } \lambda_1 \leq \lambda_2 < 0 \\ \text{unstable: } 0 < \lambda_1 \leq \lambda_2 \end{array} \right.$

→ Any saddle is unstable.

→ A focus can be $\left\{ \begin{array}{l} \text{asymptotically stable: when } \alpha < 0 \\ \text{unstable: when } \alpha > 0 \end{array} \right.$

→ Any center is stable (but not asymptotically stable).

! Theorem !

Let $m=2$, $f \in C^2(\mathbb{R}^2, \mathbb{R}^2)$ and η^* be a hyperbolic eq. point of $\dot{x} = f(x)$. The η^* is a node/saddle/focus iff for the linear system $\dot{x} = J_f(\eta^*)x$, the origin has the same type.

~ First Integrals ~

We consider the planar autonomous system:

$$(9) \quad \begin{aligned} \dot{x} &= f_1(x, y) \\ \dot{y} &= f_2(x, y), \end{aligned}$$

where $f = (f_1, f_2): \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a given C^1 function.

As usual, denote its local flow by $\varphi: \{(t, \eta) : t \in J_\eta, \eta \in \mathbb{R}^2\} \rightarrow \mathbb{R}^2$.

Def Let $U \subset \mathbb{R}^2$ be an open nonempty set. We say that $H: U \rightarrow \mathbb{R}$ is a first integral in U of (9) if it is a non-constant C^1 function and

$$H(\varphi(t, \eta)) = H(\eta), \quad \eta \in U, t \in J_\eta.$$

- A first integral in $U = \mathbb{R}^2$ is said to be a global first integral.

! A nonconstant C^1 function $H: U \rightarrow \mathbb{R}$ is a first integral in U of (9) iff it satisfies the first order linear partial differential equation:

$$(10) \quad \underline{f_1(x, y) \frac{\partial H}{\partial x}(x, y) + f_2(x, y) \frac{\partial H}{\partial y}(x, y) = 0, \quad \forall (x, y) \in U.}$$

! A first integral of the planar system (9) can be found by integrating the equation

$$\boxed{\frac{dy}{dx} = \frac{f_2(x, y)}{f_1(x, y)}}$$

cartesian differential equation of the orbits of (9)

Introduction to numerical methods

! Euler Numerical Method !

Step Size Constant $\Rightarrow$ we have the following formulas

$$\left\{ \begin{array}{l} \text{Let } h = \frac{x^* - x_0}{n} \text{ and } x_{k+1} = x_k + h \\ y_{k+1} = y_k + h \cdot g(x_k, y_k) \end{array} \right. !$$

First Order and Second Order Linear Difference Equations with Constant Coefficients

1. First Order

$$(1) \quad x_{k+1} + a x_k = b_k, \quad k \in \mathbb{N}$$

A solution of (1) is a real sequence $(x_k)_{k \geq 0}$, $x_0, x_1, \dots$, with $a \in \mathbb{R}$ a given coefficient.

$(b_k)_{k \geq 0}$ is the nonhomogeneous part (given).

! The general solution of the LHD $x_{k+1} + a x_k = 0$

can be found following the steps:

$$\text{Step 1: } \lambda + a = 0 \\ \lambda = -a$$

$$\text{Step 2: } -a \in \mathbb{R} \mapsto (-a)^k$$

$$\text{Step 3: } x_k = c \cdot (-a)^k, \quad c \in \mathbb{R}$$

! Let x_h be the gen. sol. of $x_{k+1} + a x_k = 0$ and x_p be a particular sol. of (1).

Then the gen. sol. of (1) is

$$x = x_h + x_p.$$

2. Second Order

$$(2) \quad x_{k+2} + ax_{k+1} + bx_k = 0, \quad k \in \mathbb{N}$$

A solution is a real sequence $(x_k)_{k \geq 0}$.

The characteristic equation method to find the gen. sol.:

Step 1: $r^2 + ar + b = 0 \rightarrow$ solve the eq. and find its roots $(r_{1,2} \in \mathbb{C})$

Step 2: we associate 2 sequences, following the rules.

- $\rightarrow$ if both roots are real, $r_1 \neq r_2$, then we associate $(r_1)^k$ and $(r_2)^k$
- $\rightarrow$ if $r_1 = r_2 \in \mathbb{R}$, then we associate $(r_1)^k$ and $k \cdot (r_1)^k$
- $\rightarrow$ if $r_{1,2} = \alpha \pm i\beta$, then we associate
 $\alpha, \beta \in \mathbb{R}$
 $\beta \neq 0$.
 $\operatorname{Re}[(\alpha + i\beta)^k]$ and $\operatorname{Im}[(\alpha + i\beta)^k]$

- the trigonometric form •

$$(\alpha + i\beta)^k$$

$$\alpha + i\beta = \rho(\cos \theta + i \sin \theta)$$

$$\rho = \sqrt{\alpha^2 + \beta^2}$$

$$\cos \theta = \frac{\alpha}{\rho}$$

$$\sin \theta = \frac{\beta}{\rho}$$

Step 3: We write the gen. solution $x = c_1 x^1 + c_2 x^2$;
 $c_1, c_2 \in \mathbb{R}$ and $x^1, x^2: \mathbb{N} \rightarrow \mathbb{R}$ are the sequences found at step 2.